

HRMS, Attendance, Payroll & Accounts System

Business Requirements, Workflow Logic, Data Model and Supabase Migration Specification

Prepared for Legit Design Studio

Current platform: Google Apps Script + Google Sheets Target platform: React/TanStack frontend + Supabase PostgreSQL/Auth/Storage/Edge Functions

Document version 1.0 | July 2026

Document Control

Revision History

Approvals

Table of Contents

1. Executive Summary
2. Business Context and Objectives
3. Scope and Boundaries
4. Actors and Access Model
5. End-to-End Business Workflows
6. Detailed Business Rules
7. Functional Requirements
8. Target Supabase Architecture
9. Target Data Model
10. Row Level Security and Authorization
11. API and Service Contract
12. Migration Strategy
13. Validation, Testing and Acceptance
14. Operations, Audit and Security
15. Cutover and Rollback
16. Implementation Backlog

Appendices

1. Executive Summary

This document is the authoritative functional and technical baseline for converting the existing Legit Design Studio HRMS from Google Apps Script and Google Sheets to Supabase. It consolidates the current working behavior, approved business decisions,

employee and admin workflows, payroll logic, accounts logic, security model, data structures, migration controls, test scenarios and acceptance criteria.

Admin portal manages employees, attendance, corrections, leave approvals, payroll, accounts, holidays, special working days, reports and settings.

Employee portal allows the authenticated employee to access only their own attendance, breaks, leave requests, payroll, documents, applications and profile.

Users and Employees remain separate entities: Users controls authentication and authorization; Employees contains HR and payroll profile data.

Night-shift attendance is assigned to the shift start date, covering 6:00 PM through 3:00 AM the following calendar day.

All data movement must be auditable, idempotent and reconcilable against Google Sheets totals before production cutover.

2. Business Context and Objectives

2.1 Current State

The current solution uses a React/TanStack frontend, Google Apps Script web APIs and Google Sheets as the operational database. The system is deployed and working for core employee, attendance, leave, payroll and accounts functions. The target is to replace the spreadsheet-backed persistence and script execution layer with Supabase while retaining the existing frontend experience and improving authentication, security, concurrency, auditability and maintainability.

2.2 Objectives

Provide reliable multi-user access without spreadsheet locking or row parsing risks.

Implement Supabase Auth and role-aware sessions for Admin and Employee users.

Enforce data isolation using PostgreSQL Row Level Security.

Preserve exact approved attendance, leave, payroll and profit calculations.

Support future expansion beyond the initial six employees and three partners.

Provide structured audit logs, migration reconciliation and production monitoring.

Move employee documents from ad hoc references to managed Supabase Storage while retaining optional Google Drive migration links.

2.3 Success Measures

3. Scope and Boundaries

3.1 In Scope

Authentication, user accounts, roles and sessions.

Employee profiles, employment status, compensation and document metadata.

Attendance check-in, break start, break end, checkout, geofence checks, late and deficit calculations.

Attendance correction requests and admin approval/rejection.

Leave applications, balances, paid/unpaid split and approval workflow.

Monthly attendance ledger, carry-forward deficit and payroll generation/recalculation.

Payroll status, payment status and payslip views.

Revenue, expenses, reserve ledger, profit distribution and monthly reporting.

Public holidays and special working days for all or selected employees.

Employee self-service dashboard, attendance, leave, payroll, documents, applications and profile.

Admin dashboards, reports, settings and audit logs.

Migration from current Google Sheets and Apps Script behavior to Supabase.

3.2 Out of Scope for Initial Cutover

Biometric device integrations.

Automated bank salary disbursement.

Government tax filing and EOBI integrations.

Native mobile applications.

Complex multi-company or multi-tenant billing.

Automatic Google Calendar synchronization unless explicitly retained as a post-cutover integration.

4. Actors and Access Model

4.1 Users versus Employees

5. End-to-End Business Workflows

5.1 Authentication and Routing

User enters username/email and password.

Supabase Auth validates credentials; application loads the linked application user profile.

If the user or linked employee is inactive, access is denied.

Admin is routed to /dashboard; Employee is routed to /employee.

Employee pages derive employee_id from the authenticated session, never from a frontend constant or user-editable input.

Privileged actions are additionally enforced by RLS and server-side functions.

5.2 Employee Attendance Workflow

Employee opens My Attendance and presses Check In.

Browser captures latitude/longitude; backend calculates distance from office coordinates.

Backend resolves the shift attendance date. From midnight through 2:59:59 AM, the attendance date is the previous calendar day for the 6 PM–3 AM shift.

Backend validates active employee, radius, duplicate attendance, holiday and weekend/special-working-day eligibility.

Backend stores check-in, late minutes and initial attendance status.

Employee may Start Break once, then End Break once. Break duration is calculated.

Employee must end an active break before checkout.

Checkout calculates net working minutes, deficit, late values and status. The row remains attributed to the original shift date.

Frontend refreshes the employee record and history after each successful action.

5.3 Leave Workflow

Employee selects leave type, start/end dates and reason.

System calculates inclusive total days and validates dates.

Request is created as Pending under the authenticated employee.

Admin reviews and approves or rejects.

On approval, paid balance is consumed for eligible permanent employees; any excess becomes unpaid leave.

Approved unpaid days flow into payroll deduction for the relevant month.

Rejected requests do not consume balances.

5.4 Payroll Workflow

Admin generates or recalculates the monthly attendance ledger.

System aggregates work, deficit, late counts and carry-forward minutes.

System calculates sandwich deductions and approved unpaid leave days.

Gross salary is calculated from basic salary plus eligible allowances.

Deduction days are converted to amount using gross salary divided by 30.

Bonus is added and net salary is calculated.

Admin reviews, marks payroll Paid or cancels/recalculates as permitted.

Employee sees only their own payslip and payroll history.

5.5 Accounts and Profit Workflow

Admin records revenue and expenses.

Partner salaries may be recorded as Salary-category expenses.

Net profit is calculated from revenue less expenses according to the accounts implementation baseline. Payroll treatment must not be double-counted when salary is already recorded as an expense.

30% of positive net profit is reserved and carried forward.

Each of three partners receives a mandatory 20% share of positive net profit.

Remaining 10% is retained unless business owners amend the distribution rule.

Reserve withdrawals are recorded as Reserve-category expenses and a debit in Reserve Ledger.

Monthly and overall cards are switchable; monthly values are derived from a selected YYYY-MM period.

6. Detailed Business Rules

6.1 Organization and Work Schedule

6.2 Shift Date Attribution

Check-ins from 6:00 PM onward belong to that calendar date.

Actions from 12:00 AM up to but not including 3:00 AM belong to the previous calendar date.

Checkout, break start and break end must update the active shift record rather than create a new date record.

All date comparisons must use Asia/Karachi; stored timestamps should be UTC timestampz.

6.3 Geofence Rules

Check-in is accepted only when the measured distance is within 200 meters, unless geofencing is disabled by an authorized setting.

Latitude and longitude are mandatory when geofencing is enabled.

Distance is calculated server-side using the Haversine formula or PostGIS equivalent.

Client-provided distance must never be trusted.

Because the current business has six employees, location proof is accepted on trust beyond browser geolocation; anti-spoofing is not required in initial scope.

6.4 Late Rules

An employee is late when check-in exceeds the configured start/grace threshold.

Late and deficit both apply; working late to compensate does not erase the late event. Employee must also compensate late working time on the same shift where possible.

Two late occurrences produce a half-day deduction.

Three late occurrences produce a full-day deduction.

Late counter resets after conversion at the configured threshold; remaining count is carried within the month as designed.

6.5 Break and Deficit Rules

One break interval is supported in the current baseline: break start followed by break end.

Break duration up to one hour is excluded from working minutes.

An employee may take only 30 minutes break and use the remaining 30 minutes to reduce deficit by working longer.

The target is eight net working hours.

Four deficit hours convert to a half-day deduction.

Eight deficit hours convert to a full-day deduction.

Half-day attendance does not accrue additional deficit.

Deficit and late deductions may both apply to the same payroll month.

6.6 Half-Day Rule

Current baseline marks attendance as Half Day when calculated working minutes are six hours or less.

Half Day sets deficit to zero because the half-day attendance deduction already represents the shortfall.

The threshold must be configurable before production if business owners require a different cutoff.

6.7 Weekend, Holiday and Special Working Day

Saturday and Sunday are off by default.

Public/religious/company holidays are paid holidays when entered by Admin.

Admin creates holiday records manually unless an external calendar integration is retained.

Admin can create a special working day for all employees or a selected subset.

Only assigned employees may mark attendance on a special working day.

A special working day must explicitly override weekend/holiday restrictions for the assigned employee set.

6.8 Leave Rules

Permanent leave balances reset based on permanent date anniversary, e.g., 1 April.

Approved paid leave consumes the corresponding balance.

Approved excess or explicitly Unpaid leave becomes unpaidDays.

Unpaid leave is deducted using monthly payroll daily rate.

Legacy requests with blank totalDays must be preserved and flagged for migration cleansing rather than silently recalculated without review.

6.9 Sandwich Policy

Friday and Monday absence around a normal weekend causes weekend days to be deducted.

Friday-only or Monday-only absence applies the approved half-sandwich treatment.

Policy calculation must exclude holidays and special working day exceptions appropriately.

Sandwich days are included in payroll deduction days.

6.10 Payroll Formula

Permanent employees receive fuel and OPD allowances; other employee types receive them only if configured.

Payroll requires an attendance ledger for the employee and month.

Recalculation must update or version the existing record; duplicate active payroll for the same employee/month is prohibited.

Carry-forward deficit follows Option B: unresolved minutes carry to the next month.

Marking Paid records paid_at and prevents ordinary recalculation unless an authorized reversal is performed.

6.11 Accounts and Reserve

Net profit = revenue – expenses, provided payroll/salary is represented only once.

30% reserve carries forward.

Partner profit share is mandatory: 20% each for three partners.

Partner salary is separate from partner profit share and may be recorded through Salary expense.

Reserve withdrawal is recorded as an expense and reserve ledger debit.

Reserve balance is derived from ledger transactions, not manually overwritten.

7. Functional Requirements

8. Target Supabase Architecture

8.1 Logical Components

8.2 Design Principles

Use UUID primary keys and timestampz timestamps.

Use date for attendance_date and month_start date for monthly records.

Use numeric(14,2) for monetary values and integer minutes for durations.

Use PostgreSQL enums or constrained text for stable statuses.

Use database constraints to prevent duplicate employee/month and employee/date records.

Put security-critical identity derivation and calculations on the server/database side.

Use views for dashboards; avoid repeatedly downloading every row to calculate totals in the browser.

9. Target Data Model

profiles

employees

attendance

attendance_corrections

leave_balances

leave_requests

attendance_ledgers

payroll

revenues

expenses

reserve_ledger

holidays

special_working_days

special_working_day_employees

employee_documents

employee_applications

audit_logs

9.1 Mandatory Constraints and Indexes

Unique employees.employee_code.

Unique profiles.username where username is used.

Unique attendance(employee_id, attendance_date).

Unique attendance_ledgers(employee_id, month_start).

Unique active payroll(employee_id, month_start) or versioned payroll with one effective record.

Unique holidays.holiday_date unless multiple holiday labels per date are intentionally supported.

Indexes on attendance(employee_id, attendance_date desc), leave_requests(employee_id, created_at desc), payroll(employee_id, month_start desc), revenue/expense date and audit entity fields.

Check constraints for non-negative minute fields, amounts and percentage-compatible days.

10. Row Level Security and Authorization

11. API and Service Contract

11.1 Recommended RPC/Edge Functions

11.2 State Machine Requirements

12. Migration Strategy

12.1 Source Sheets

Users, Employees, Attendance, Attendance Ledger, Attendance Corrections, Leave Requests, Leave Balances, Payroll, Revenue, Expenses, Reserve Ledger, Holidays, Special Working Days and Settings.

Exact sheet names and headers must be captured before extraction. Existing examples include Reserve Ledger headers: reserveld, transactionDate, transactionType, amount, balanceAfter, description, createdAt.

12.2 Migration Phases

Freeze schema changes in Apps Script/Sheets and export a versioned source snapshot.

Create Supabase schema, enums, constraints, RLS, storage buckets and functions in a non-production project.

Extract each sheet to CSV/JSON using a deterministic export script.

Normalize dates, booleans, UUIDs, blank numeric fields and employee/user relationships.

Load reference/master data first: employees, profiles/users mapping, settings, holidays.

Load transactional data in dependency order: attendance, corrections, leave, ledgers, payroll, accounts, reserve, documents metadata.

Reconcile row counts, employee-level totals, monthly totals and payroll outputs.

Run parallel read-only comparison and workflow UAT.

Perform final delta migration during a controlled write freeze.

Switch frontend environment variables to Supabase and monitor.

Retain source Sheets read-only for the agreed retention period.

12.3 Data Transformation Rules

12.4 Reconciliation Checks

Row counts per sheet/table.

Employees and user-to-employee link coverage.

Attendance rows per employee/month and total working/deficit minutes.

Leave paid/unpaid totals and remaining balances.

Payroll gross, deduction and net totals per employee/month.

Revenue, expense and reserve balances by month and overall.

Holiday and special-day dates/assignments.

Invalid or orphaned foreign keys; none may remain unresolved at go-live.

13. Validation, Testing and Acceptance

13.1 Critical UAT Scenarios

13.2 Non-Functional Acceptance

No critical console/runtime errors on admin or employee routes.

Queries return empty arrays/states without page crashes.

All mutation buttons show pending state, success refresh and clear error message.

Core pages are usable on desktop and mobile widths.

No secrets or service-role key are exposed in browser code.

Backup and rollback are tested before go-live.

14. Operations, Audit and Security

Use separate development, staging and production Supabase projects or schemas.

All schema changes are migration files committed to version control.

Service-role key is server-only.

Enable point-in-time recovery/backups appropriate to the plan.

Capture audit entries for employee changes, leave decisions, corrections, payroll actions, settings and account transactions.

Use signed URLs for private employee documents.

Log failed authentication and privileged action failures.

Monitor RPC failures, database errors and migration anomalies.

14.1 Settings to Seed

15. Cutover and Rollback

15.1 Cutover Checklist

Production schema migrations applied and verified.

RLS test suite passed using Admin and Employee accounts.

Final source backup and export completed.

Delta migration completed and reconciled.

User invitations/password reset process completed.

Frontend production environment points to production Supabase.

DNS/hosting deployment validated.

Smoke test: login, check-in, break, checkout, leave, payroll read, accounts read.

Business owner approval recorded.

15.2 Rollback

Keep Apps Script/Sheets available in read-only or controlled fallback mode during stabilization.

If a critical failure occurs before new writes become authoritative, restore frontend to Apps Script endpoint.

If writes occurred in Supabase, export delta transactions before rollback and define manual reconciliation ownership.

Never allow both platforms to accept unrestricted writes simultaneously after cutover.

16. Implementation Backlog

Appendix A — Approved Business Decisions

Half day is already counted; no deficit is added for a half-day attendance record.

Public holidays are manually added by Admin and are paid holidays.

Unpaid leave is deducted in payroll.

Office geofence radius is 200 meters.

Basic salary and allowances form gross salary.

Late and deficit both apply.

Employees may work later to compensate hours, but the late event remains.

Special working days may apply to all employees or selected employees only.

Attendance belongs to the date on which the shift/check-in started.

Partner salary may be recorded as an expense; mandatory 20% profit share remains separate.

CNIC, educational certificates and medical documents are managed as employee documents.

Deficit carry-forward uses Option B: carry hours/minutes to the next month.

Employee location is trusted at the current team size; no advanced anti-spoofing requirement.

Appendix B — Migration Issue Register Template

Appendix C — Definition of Done

Implementation matches this specification or has an approved change request.

Database migration is versioned and reproducible.

RLS and role tests pass.

Critical workflow and payroll tests pass.

Migration reconciliation is signed off.

Production monitoring, backup and rollback are documented.

Admin and employee user guides are updated.

Apps Script/Sheets are archived according to the cutover plan.